Constraint-Aware Service Placement for Privacy-Preserving Edge Computing Using Safe Reinforcement Learning

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안전 강화학습을 활용한 프라이버시 보존 엣지 컴퓨팅을 위한 제약 인지형 서비스 배치

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Abstract

This paper studies RL for privacy-preserving service placement in edge computing. The objective is to maximize average resource utilization while satisfying capacity and privacy-conflict constraints. We formulate the problem as an ILP baseline and compare it with several Safe RL methods under three ECU-to-service ratio settings. Experimental results show that constraint-aware PPO methods, especially Maskable-PPO, achieve utilization close to the ILP baseline with zero or near-zero constraint violations, whereas unconstrained PPO produces inflated utilization by violating privacy-conflict constraints.

I. Introduction

Edge computing reduces latency by placing computation closer to users and data sources, but limited ECU resources make service placement a constrained allocation problem. Traditional methods such as ILP can solve this problem when objectives and constraints are clearly defined, but complex deployment environments require increasingly detailed formulations, especially when resource and privacy constraints must be considered together.

Privacy conflicts may arise from different permission requirements, data access scopes, or privacy policies, so conflicting services should not be placed on the same ECU. Although RL can learn placement policies through interaction with the environment, standard RL methods do not guarantee constraint satisfaction.

This paper investigates constraint-aware RL for privacy-preserving service placement in edge computing. We formulate average resource utilization as the objective and capacity and privacy-conflict constraints as feasibility requirements, then compare an ILP baseline with several RL methods under three ECU-to-service ratio settings. The results show that constraint-aware PPO methods can approximate ILP performance while maintaining feasible deployment results.

II. Methods

A. System Model and Problem Formulation

We consider a service placement problem with N edge computing units (ECUs) and M services. Each ECU n has a limited resource capacity c_n, and each service m requires e_m virtual machines. The binary decision variable x_{nm} indicates whether service m is assigned to ECU n, and y_n indicates whether ECU n is active. The resource utilization of ECU n is defined as: $R_n = \frac{1}{c_n}\sum_m e_m x_{mn}$. The average resource utilization is calculated over all active ECUs: $AR = \frac{1}{\sum_n y_n}\sum_n R_n$.

The objective is to maximize AR while satisfying capacity and privacy-conflict constraints. The capacity constraint requires that the total VM demand assigned to each ECU does not exceed its capacity: $\sum_m e_m x_{mn} \leq c_n, \forall n$.

Privacy conflict is modeled as an incompatibility relation between services. If two services i and j have a privacy conflict, they cannot be deployed on the same ECU: $x_n i + x_n j \leq 1, \forall n, (i,j) \in P$, where P denotes the set of service pairs with privacy conflicts. These constraints ensure that each placement result is both resource-feasible and privacy-preserving.

B. Proposed Idea

We use an ILP-based optimization method as the reference baseline and compare it with several RL methods. The ILP formulation solves the service placement problem as a constrained optimization problem, and the Dinkelbach algorithm is used to handle the fractional average-utilization objective.

The compared RL methods include DQN, DDQN, standard PPO, Maskable-PPO, Lagrangian-PPO, and PPO with repair. Standard PPO is evaluated as an unconstrained policy optimization method. Maskable-PPO prevents invalid actions during policy learning

through action masking. Lagrangian-PPO introduces constraint-related penalties into the learning objective, while PPO with repair corrects infeasible placement results after action selection. DQN and DDQN are included as value-based RL baselines.

All methods are evaluated under the same randomly generated placement scenarios. The goal is to achieve high average resource utilization while keeping both capacity and privacy-conflict violations at zero or near zero.

C. Experiments

The experiments are designed to evaluate how different methods perform under different ECU-to-service ratios. The three variants are

Table 1. Experimental scenarios based on ECU-to-service relationship

Label	ECU-to-service relationship
LT	$N < M$
EQ	$N = M$
GT	$N > M$

In the LT setting, the number of ECUs is smaller than the number of services. This setting is the most constrained because multiple services must be placed on the same ECU more frequently. In the EQ setting, the number of ECUs equals the number of services. This setting represents a balanced deployment condition. In the GT setting, the number of ECUs is larger than the number of services, making it easier to separate conflicting services. The specific experimental settings are

Table 2. Experimental settings for ECU and service numbers

Label	Number of ECUs	Number of services
LT	10	15
EQ	10	10
GT	15	10

For each setting, random placement scenarios are generated and shared across all compared methods. In the EQ and GT settings, 40 held-out test scenarios are used for evaluation. In the LT setting, infeasible scenarios are excluded in advance, and 32 feasible test scenarios are used for evaluation.

The evaluation metrics are average resource utilization, total constraint violations, capacity violations, and privacy-conflict violations. A method is considered effective only when it achieves high average resource utilization while keeping constraint violations at zero or near zero.

The ILP baseline achieves average resource utilization values of 0.751, 0.548, and 0.626 in the LT, EQ, and GT settings, respectively. These values are used as reference results for comparing the RL methods.

Among the RL methods, Maskable-PPO, Lagrangian-PPO, and PPO with repair achieve average resource utilization close to the ILP baseline. More importantly, these methods maintain zero or near-zero constraint violations across the tested settings. This shows that constraint-aware PPO methods can learn feasible placement policies while preserving strong resource utilization performance.

III. Conclusion

This paper studied constraint-aware RL for privacy-preserving service placement in edge computing, where average resource utilization was optimized under capacity and privacy-conflict feasibility constraints. Using an ILP-based method as the reference baseline, several RL methods were evaluated under LT, EQ, and GT ECU-to-service ratio settings. The results show that constraint-aware PPO methods, especially Maskable-PPO, achieve resource utilization close to the ILP baseline while maintaining zero or near-zero constraint violations, whereas unconstrained PPO produces misleadingly high utilization by violating privacy-conflict constraints. These findings confirm that resource utilization is meaningful only when feasibility constraints are satisfied, and indicate that Maskable-PPO offers the best balance between constraint satisfaction and resource utilization among the tested methods.

ACKNOWLEDGMENT

This work was partly supported by the IITP(Institute of Information & Communications Technology Planning & Evaluation)-ITRC(Information Technology Research Center) grant funded by the Korea government(Ministry of Science and ICT)(IITP-2026-RS-2023-00259967, 50%) and 「Fostering Project of R&D promotion complex in Busan」 funded by the Korea government(MSIT) and a local government(Busan)(2710089900, 50%).

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